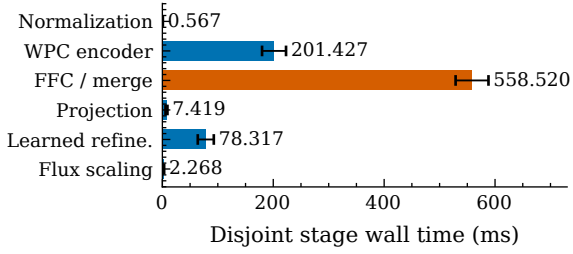
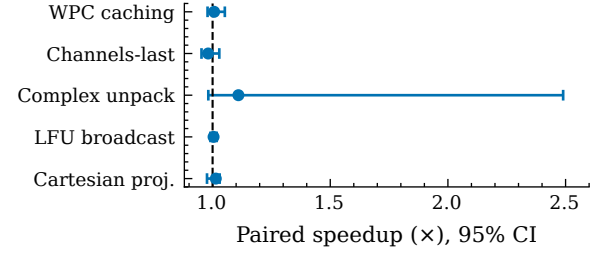
Deep phase retrieval on CPU: execution, optimization and reliability

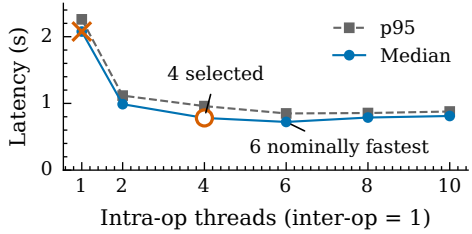
(a) Disjoint profile: mean ± SD, five passes



(b) Exact candidates; none promoted



(c) Sequential thread screen; 36 calls / setting

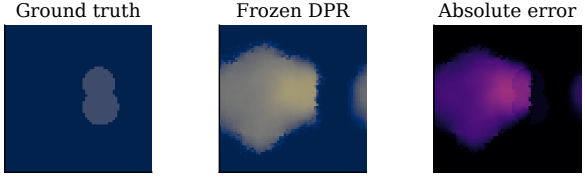


(d) Representative high-photon case



b0012__corrected_p10000000_b16
Aligned NRMSE 0.037; SSIM 0.967

(e) Locked worst case: OOD, low photons



b0016__legacy_p10000_b48
Aligned NRMSE 9.652; SSIM 0.348

(f) Absolute reference reliability

Metric	Base mean	95% base CI
Aligned NRMSE	1.498	[1.462, 1.540]
SSIM	0.739	[0.693, 0.776]
Failure fraction	0.440	[0.402, 0.507]

251 / 570 failures (aligned NRMSE > 0.5)
15 independent objects; 38 conditions / object
10 test + 5 OOD objects; synthetic data
Selected: original FP32 DPR1, 4 / 1 threads

